

Supplementary Material

Alternative Structural Measures and Ridge Robustness

The first three figures reproduce the main structural-advantage analyses with the broker's two other saved ego-network measures, Burt's aggregate network constraint and Burt's effective size. Both are recomputed for the broker every 20 periods. Figure S4 reports the focused paired-Ridge robustness analysis discussed in the main text. As in the main text, a regime is one scientifically distinct parameter configuration; duplicate grid coordinates and inactive parameters do not create additional regimes.

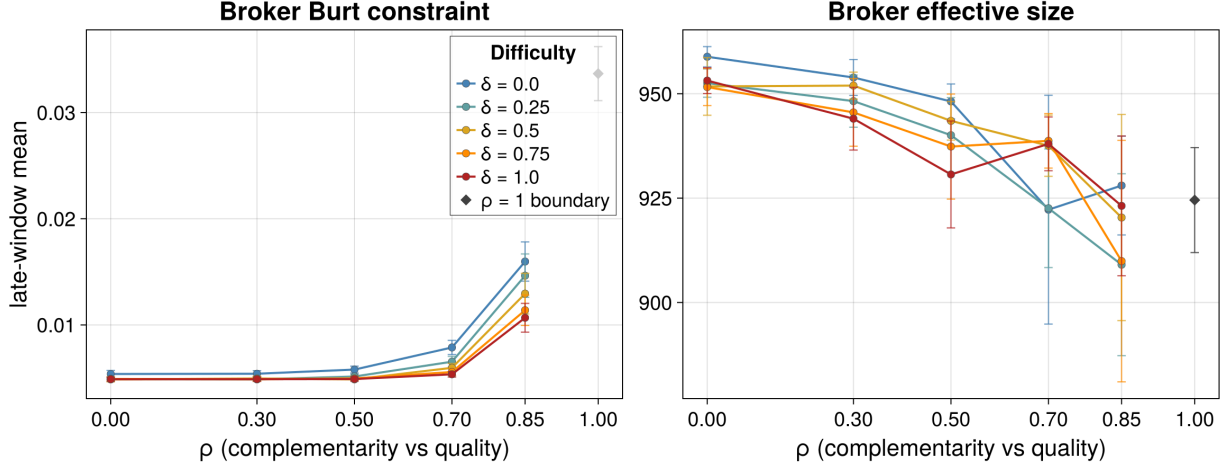


Figure S1: Broker Burt constraint (left) and broker effective size (right) across the $\rho \times \delta$ grid. Lines for each difficulty level δ span $\rho = 0$ through 0.85. The pure-quality boundary $\rho = 1$, where δ is inactive, is shown once and unconnected. Markers are late-window means over seeds and whiskers are 95% Monte Carlo intervals.

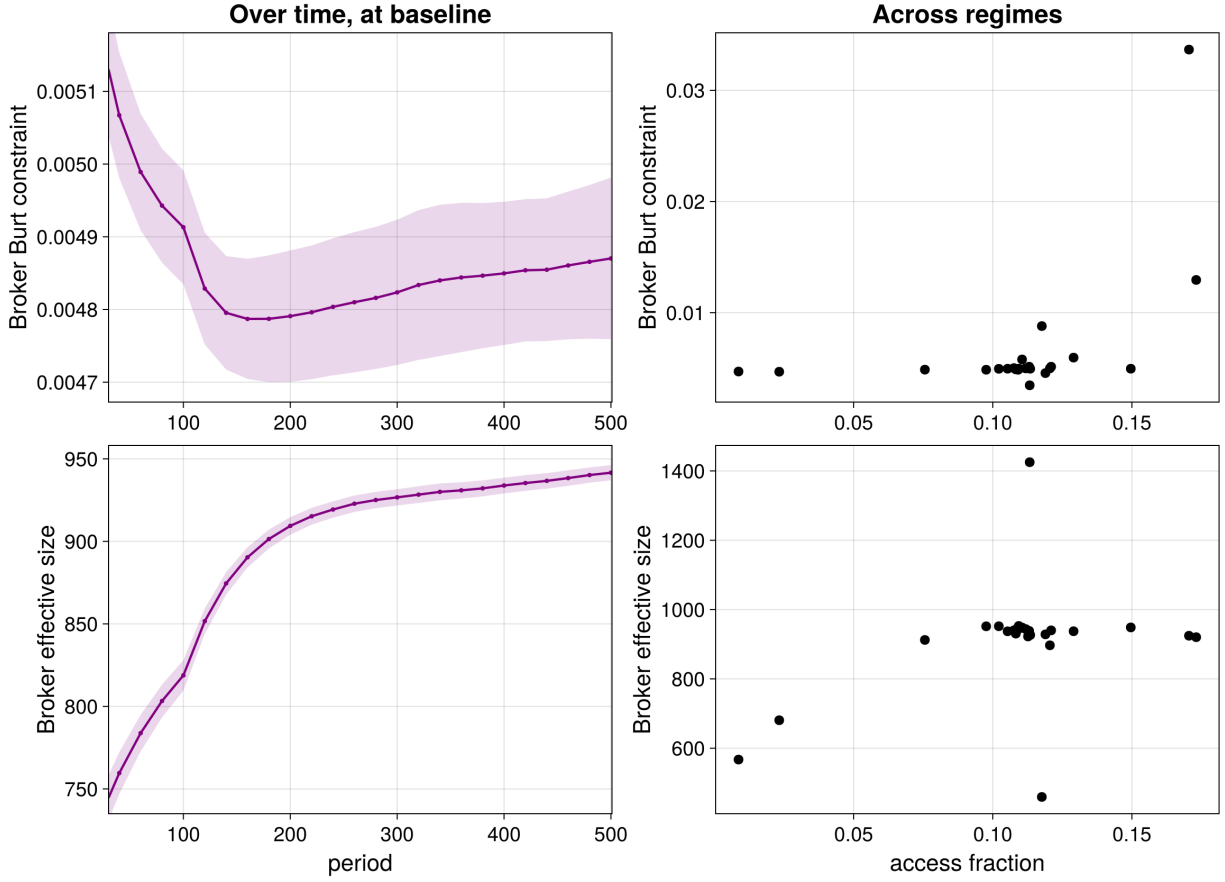


Figure S2: Broker Burt constraint (top) and broker effective size (bottom). Left: the measure over time at the baseline, as 5-observation rolling ensemble means over seeds, with pointwise 95% Monte Carlo intervals, measured every 20 periods. Right: one dot per one-at-a-time regime, the measure against access fraction, each a late-window mean over that regime's seeds. Time starts at $t = 30$ to omit the initialization transient.

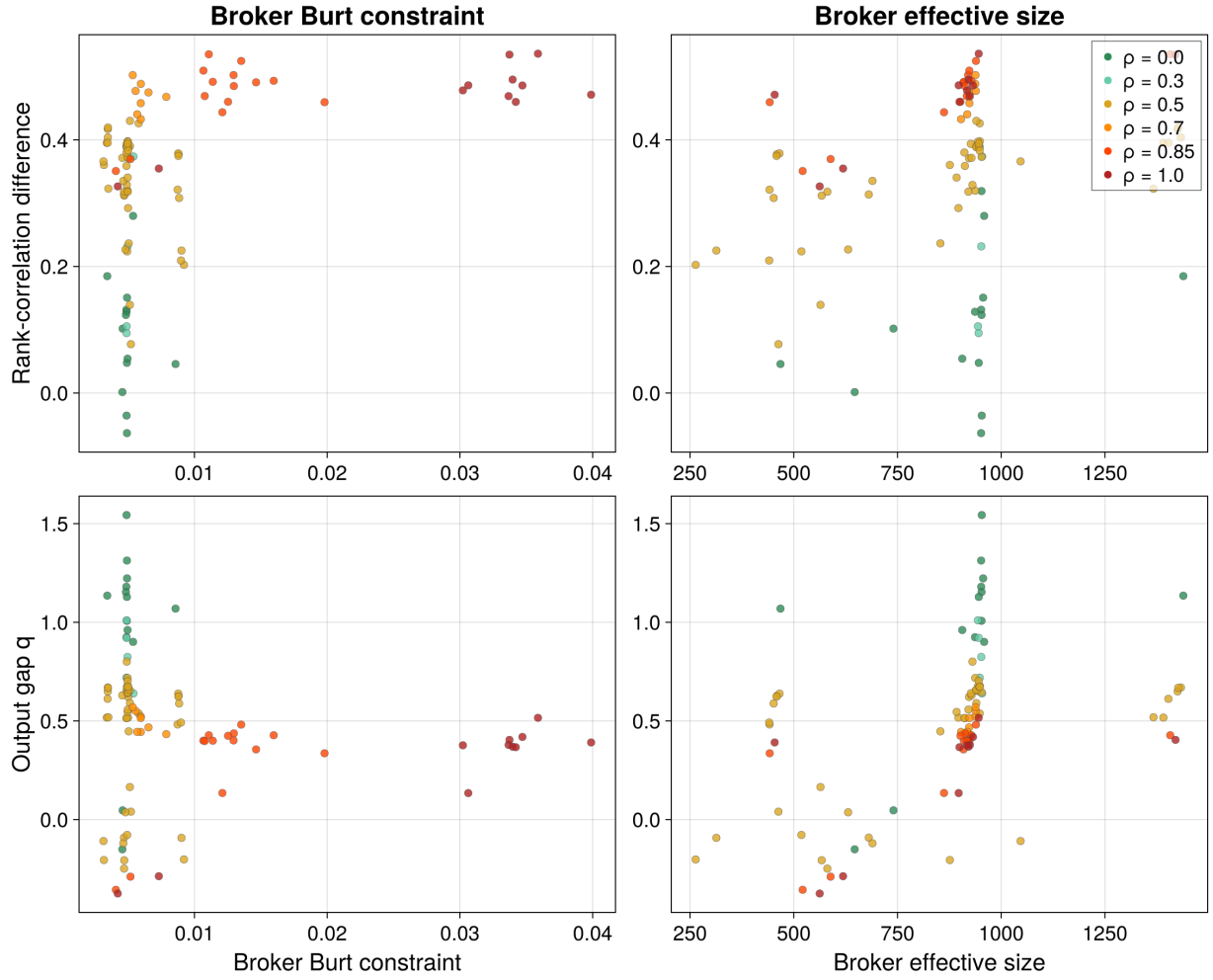


Figure S3: Difference in holdout rank correlation (broker minus principal, top) and output gap (broker channel minus self-search, bottom) against broker Burt constraint (left) and broker effective size (right); one dot per regime, each a late-window mean over that regime's seeds, colored by the matching parameter ρ .

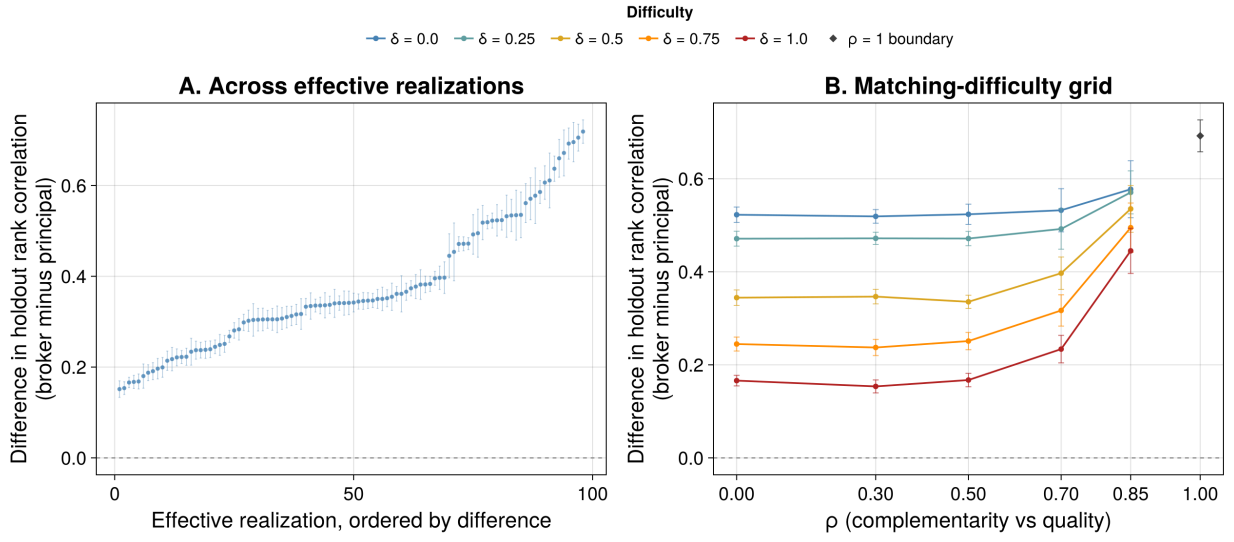


Figure S4: Broker ranking advantage under paired-Ridge learning. Panel A orders the 98 regimes by their late-period broker-minus-principal difference in holdout rank correlation; points are regime means and whiskers are 95% Monte Carlo intervals over seeds. Panel B shows the same difference across the $\rho \times \delta$ design, with one line per value of δ over $\rho = 0$ through 0.85. The pure-quality boundary $\rho = 1$, where δ is inactive, is shown once and unconnected. Markers are regime means, whiskers are 95% Monte Carlo intervals, and the horizontal line marks equal broker and principal holdout rank correlations.